

Lucian Medrihan

Senior Full Stack Engineer

About Me

Passionate Software Engineer with Full Stack development experience, which began by exploring Ethical Hacking at age 12. My career spans from GIS applications to management systems, with a focus on machine learning projects. I have contributed to projects for SIAN and ESA, demonstrating strong problem-solving capabilities.

Contact

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Rome, Italy

Skills

Technical Skills: Proficient in **Python**, JavaScript, and Node.js, with extensive experience in **React.js**, **Angular.js**, Flask, and FastAPI. Expertise in **full-stack development** using modern web technologies, complemented by strong skills in **Docker**, **AWS**, Cloudfront, and cloud solutions.

Additional Skills: **GIS**, **Machine Learning**, **LLMs**, Image Processing, Tailwind CSS, Bootstrap, Openlayers, Linux commands, **Cybersecurity**, Geopandas, JQuery, network programming.

Soft Skills: Proven ability to solve **complex problems**, manage projects effectively, and **lead teams**. Strong time management, critical thinking, and attention to detail.

Education

● **Computer and Automation Engineering**
Sapienza University of Rome
Graduated: 03/2025

Languages

Italian	★ ★ ★ ★ ★
Romanian	★ ★ ★ ★ ★
English	★ ★ ★ ★ ★

Achievements & Activities

- **Bug Bounty Hunter:** Found 2 critical RCEs (CVSS 10.0) on major Italian ISP via Cyberdart.eu (\$5k bounties)
- **Google Workshop Winner** for UniCity, presented at Google Tel Aviv
- Member of **Hackappatoi**, Sapienza's top ethical hacking team

Experiences

● **Full Stack Software Engineer**
PR.EL.CART. EUROPE ENGINEERING S.R.L.
Rome, Italy
Jun 2025 - Present

- Developing advanced software solutions in collaboration with the Italian Ministry of Agriculture
- Managing and mentoring a team of 3 junior developers, guiding them in software architecture
- Independently developed the official AGEA Vinitaly 2026 application using PMTiles technology
- Engineered an AI pipeline on Google Cloud Vertex AI to generate 1m satellite photos from 10m Sentinel imagery, reducing costs from €4/5M to €500/month

● **Full Stack Software Engineer**
Tiuke S.R.L.
Rome, Italy
Oct 2019 - Jun 2025

- Developed web and mobile applications for the Italian Ministry of Agriculture
- Led development teams on multiple web applications, managing task assignments and coordinating colleagues' work
- Optimized legacy applications into responsive web applications
- Created internal automation tools for pre-configured layers with API services
- Developed backend system for automatic documentation generation
- Implemented LLM applications with specialized agents for data extraction
- Improved geospatial segmentation models on satellite imagery
- Created batch scripts for processing large volumes of geospatial data

Projects

[View more on GitHub](#)

Sentinel AI Pipeline
Satellite image upscaling using AI
Completed

- Designed a pipeline on Google Cloud Vertex AI to obtain 1m resolution photos from 10m Sentinel-2 imagery
- Reduced operational costs from €4 million (over 3 years) to €500 per month, enabling monthly updates

Card Scanner
AI-powered Pokemon card recognition system with client-side processing
Completed

- Built a lightweight (<10MB) client-side card identification system that works with all new cards without requiring retraining
- Successfully integrated into TCG Pocket Collection Tracker, the world's most popular Pokemon TCG tracker, significantly enhancing user experience

AGEA Vinitaly 2026
Official web application for Vinitaly 2026
Completed

- Developed the application using PMTiles technology to efficiently visualize agricultural plots across all of Italy
- Managed massive geospatial datasets ensuring high rendering performance

Wolf & Mafia
100% Offline Mafia & Werewolf game for Android and iOS
Completed

- Developed from scratch a complete offline social deduction game featuring 36 unique roles and a fully guided narrator system
- Implemented QR-code based local multiplayer allowing users to join games instantly without an internet connection

Game Ecosystem Tools
ForgeMaster App & Eatventure Seed Predictor
In Progress

- Developed widely-used reverse engineering tools, including an auto-updating app for ForgeMaster and an accurate RNG drop predictor for Eatventure

Openlayers GIS Applications
Enterprise-level geospatial web applications
Completed

- Developed full-stack solutions for the Italian Ministry of Agriculture that process massive geographic datasets while maintaining high performance
- Implemented advanced features including LLM-powered data extraction and SAM-based segmentation models for agricultural applications

UniCity
Award-winning collaborative platform for university students
Completed

- Created an intelligent note-sharing system with advanced indexing algorithms, winning the Google Workshop competition
- Presented the project at Google's offices in Tel Aviv, receiving recognition for innovation and practical application

To view my digital business card, open your terminal and type: `npz lvcian`